MODULE 2: ENERGY DEVICES

(CO: 2)

Complete Study Textbook, Step-by-Step Derivations & VIT FAT Question Bank

Aligned with Official VIT BACHY101 Course Page Syllabus, Question Bank, and Past FAT Papers

Scope of This Textbook (Official VIT Syllabus - All 8 Topics): (1) Electrochemical and electrolytic cells, electrode materials with examples, (2) Nernst equation and numericals, (3) Classification of batteries, (4) Chemistry of Li-ion secondary batteries, (5) Fuel cells: $H_2 - O_2$ and solid oxide fuel cell (SOFC), (6) Solar cells: photovoltaic silicon based, (7) Photoelectrochemical cells, and (8) Dye-sensitized solar cells (DSSC).

1 VIT FAT Exam Blueprint & Master Formula Sheet

In VIT Final Assessment Tests (BACHY101 FAT), Module 2 carries **15 to 20 marks**. Typical 10-mark questions include:

1. **Question Type A (10 Marks):** Nernst equation derivation and cell potential numericals, including equilibrium constant K_{eq} and Gibbs free energy ΔG°.
2. **Question Type B (10 Marks):** Chemistry of Lithium-ion batteries (charging/discharging intercalation equations at cathode and anode) or Solid Oxide / Hydrogen-Oxygen Fuel Cell working principles.
3. **Question Type C (10 Marks):** Principle and electron transport mechanism in Dye-Sensitized Solar Cells (DSSC / Gratzel cell) or silicon photovoltaic p-n junction.

Device / Law	Governing Equation / Formula	VIT Exam Key Insight
Nernst Equation at 25° C	$E_{cell} = E^\circ_{cell} - \frac{0.0591}{n} \log_{10} Q$	$E^\circ_{cell} = E^\circ_{cathode} - E^\circ_{anode}$ (both as reduction potentials).
Thermodynamic Relations	$\Delta G = -nFE_{cell}, \quad \Delta G^\circ = -RT \ln K_{eq}$	$F = 96500$ C/mol; positive E_{cell} indicates spontaneous reaction.
Lithium-Ion Cathode	$LiCoO_2 \Rightarrow Li_{1-x}CoO_2 + xLi^+ + xe^-$	Intercalation/de-intercalation process; no metallic lithium deposition.
Lithium-Ion Anode	$xLi^+ + xe^- + 6\ C \Rightarrow Li_xC_6$	Graphite host structure prevents dendritic dendrite formation.
$H_2 - O_2$ Alkaline Fuel Cell	$E^\circ_{cell} = 1.229$ V, $\quad 2H_2 + O_2 \Rightarrow 2H_2O$	Anode: $H_2 + 2OH^- \Rightarrow 2H_2O + 2e^-$; Cathode: $O_2 + 2H_2O + 4e^- \Rightarrow 4OH^-$.
DSSC / Gratzel Cell	$\eta = \frac{P_{max}}{P_{in}} = \frac{V_{oc} \times J_{sc} \times \text{FF}}{P_{in}}$	Ruthenium dye absorbs photon, injects electron into TiO_2 conduction band.

2 Electrochemical Cells & The Nernst Equation

2.1 Electrochemical vs. Electrolytic Cells

- **Electrochemical (Galvanic / Voltaic) Cell:** Converts chemical energy into electrical energy through spontaneous redox reactions ($\Delta G < 0, E_{cell} > 0$). Anode is negative, cathode is positive.
- **Electrolytic Cell:** Converts electrical energy into chemical energy to drive non-spontaneous reactions ($\Delta G > 0$). Requires external DC power source. Anode is positive, cathode is negative.

2.2 The Nernst Equation Derivation

From thermodynamics, the Gibbs free energy change under non-standard conditions is:

$$\Delta G = \Delta G^\circ + RT \ln Q$$

Since $\Delta G = -nFE_{\text{cell}}$ and $\Delta G^\circ = -nFE^\circ_{\text{cell}}$:

$$-nFE_{\text{cell}} = -nFE^\circ_{\text{cell}} + RT \ln Q$$

Dividing throughout by $-nF$:

$$E_{\text{cell}} = E^\circ_{\text{cell}} - \frac{RT}{nF} \ln Q$$

At $T = 298.15$ K (25° C), with $R = 8.314 \frac{\text{J}}{\text{mol} \cdot \text{K}}$ and $F = 96485$ C/mol:

$$\frac{2.303RT}{F} = \frac{2.303 \times 8.314 \times 298.15}{96485} = 0.05916 \text{ V}$$

$$E_{\text{cell}} = E^\circ_{\text{cell}} - \frac{0.0591}{n} \log_{10} Q$$

At chemical equilibrium, $E_{\text{cell}} = 0$ and $Q = K_{\text{eq}}$:

$$E^\circ_{\text{cell}} = \frac{0.0591}{n} \log_{10} K_{\text{eq}} \Rightarrow \log_{10} K_{\text{eq}} = \frac{nE^\circ_{\text{cell}}}{0.0591}$$

3 Battery Systems & Secondary Lithium-Ion Batteries

3.1 Battery Classifications

1. **Primary Batteries:** Non-rechargeable electrochemical cells where reactants are irreversibly consumed (e.g., Leclanche zinc-carbon, alkaline zinc-manganese dioxide cell).
2. **Secondary (Storage) Batteries:** Rechargeable cells where cell reactions are reversed by passing electric current in the opposite direction (e.g., Lead-acid accumulator, Nickel-Cadmium, Lithium-ion battery).
3. **Reserve Batteries:** Inactive until electrolyte is added or activated by heat/shock (used in missiles and ocean buoys).

3.2 Chemistry of Lithium-Ion Secondary Batteries

A commercial Lithium-ion cell consists of:

- **Anode (Negative Electrode):** Intercalated lithium in graphite layers (Li_xC_6).
- **Cathode (Positive Electrode):** Lithium transition metal oxide, typically lithium cobalt oxide (LiCoO_2), lithium iron phosphate (LiFePO_4), or NMC ($\text{LiNi}_x\text{Mn}_y\text{Co}_z\text{O}_2$).
- **Electrolyte:** Non-aqueous lithium salt (typically LiPF_6) dissolved in organic carbonate mixtures (ethylene carbonate + dimethyl carbonate, EC + DMC).
- **Separator:** Microporous polyethylene/polypropylene membrane.

Chemical Reactions During Discharging:

- **At Anode (Oxidation):**

$$\text{Li}_x\text{C}_6 \Rightarrow 6\text{ C} + x\text{Li}^+ + xe^-$$

- **At Cathode (Reduction):**

$$\text{Li}_{1-x}\text{CoO}_2 + x\text{Li}^+ + xe^- \Rightarrow \text{LiCoO}_2$$

- **Overall Cell Discharge:**

$$\text{Li}_x\text{C}_6 + \text{Li}_{1-x}\text{CoO}_2 \Rightarrow 6\text{ C} + \text{LiCoO}_2 \quad (E_{\text{cell}} \approx 3.7 \text{ V})$$

During charging, the external voltage source drives Li^+ ions out of the cathode, through the organic electrolyte, and into the graphite layers at the anode.

4 Fuel Cells & Solar Photovoltaic Technologies

4.1 Hydrogen-Oxygen Alkaline Fuel Cell

A fuel cell converts the chemical energy of a continuously supplied fuel and oxidant directly into electricity:

- **Anode Reaction:** $2H_2(\text{g}) + 4OH^-(\text{aq}) \Rightarrow 4H_2O(\text{l}) + 4e^-$ ($E^\circ = -0.83$ V)
- **Cathode Reaction:** $O_2(\text{g}) + 2H_2O(\text{l}) + 4e^- \Rightarrow 4OH^-(\text{aq})$ ($E^\circ = +0.40$ V)
- **Overall Reaction:** $2H_2(\text{g}) + O_2(\text{g}) \Rightarrow 2H_2O(\text{l})$ ($E^\circ_{\text{cell}} = 1.23$ V)

Efficiency of fuel cell: $\eta = \frac{\Delta G}{\Delta H} \times 100\% \approx 83\%$, far superior to the Carnot limit of combustion heat engines!

4.2 Solid Oxide Fuel Cells (SOFC)

Operates at elevated temperatures (700° C − 1000° C) utilizing a dense ceramic solid electrolyte: Yttria-Stabilized Zirconia (YSZ).

- **Cathode:** Porous Lanthanum Strontium Manganite (LSM): $O_2 + 4e^- \Rightarrow 2O^{2-}$.
- Oxygen ions O^{2-} migrate through the vacant oxide sites of the YSZ crystal lattice.
- **Anode:** Nickel-YSZ cermet: $2H_2 + 2O^{2-} \Rightarrow 2H_2O + 4e^-$.

4.3 Dye-Sensitized Solar Cells (DSSC / Gratzel Cell)

1. **Dye Absorption:** A ruthenium polypyridyl dye (e.g., N719) absorbs an incoming solar photon and undergoes photoexcitation: $\text{Dye} + h\nu \Rightarrow \text{Dye}^*$.
2. **Electron Injection:** The excited dye injects an ultrafast electron into the conduction band of nanostructured titanium dioxide (TiO_2): $\text{Dye}^* + TiO_2 \Rightarrow \text{Dye}^+ + e^-(TiO_2)$.
3. **Percolation & Power:** Electrons percolate through the mesoporous TiO_2 nanoparticle network to the Fluorine-doped Tin Oxide (FTO) current collector, flowing through external load.
4. **Dye Regeneration:** The oxidized dye (Dye^+) is reduced by iodide ions in the redox electrolyte: $2\text{Dye}^+ + 3I^- \Rightarrow 2\ \text{Dye} + I_3^-$.
5. **Electrolyte Regeneration:** Triiodide (I_3^-) diffuses to the platinum counter electrode where it is reduced back to iodide: $I_3^- + 2e^- \Rightarrow 3I^-$.

5 Worked VIT Question Bank & Past FAT Problems

Problem 2.1 (VIT Question Bank Q8 - 10 Marks): Calculate the EMF of the following galvanic cell at 25° C:

$$\text{Zn(s)} \mid \text{Zn}^{2+}(0.02\text{ M}) \parallel \text{Cu}^{2+}(0.50\text{ M}) \mid \text{Cu(s)}$$

Given standard reduction potentials: $E^\circ\left(\frac{\text{Zn}^{2+}}{\text{Zn}}\right) = -0.763$ V and $E^\circ\left(\frac{\text{Cu}^{2+}}{\text{Cu}}\right) = +0.337$ V. Also calculate the maximum electrical work available per mole of copper deposited.

Solution:

1. Identify half-reactions: Anode (oxidation): $\text{Zn(s)} \Rightarrow \text{Zn}^{2+} + 2e^-$ Cathode (reduction): $\text{Cu}^{2+} + 2e^- \Rightarrow \text{Cu(s)}$ Overall cell reaction: $\text{Zn(s)} + \text{Cu}^{2+} \Rightarrow \text{Zn}^{2+} + \text{Cu(s)}$, so $n = 2$.
2. Standard cell potential:

$$E^\circ_{\text{cell}} = E^\circ_{\text{cathode}} - E^\circ_{\text{anode}} = +0.337 - (-0.763) = 1.100\text{ V}$$

3. Apply Nernst equation:

$$E_{\text{cell}} = E^\circ_{\text{cell}} - \frac{0.0591}{n}\log_{10}\left(\frac{[\text{Zn}^{2+}]}{[\text{Cu}^{2+}]}\right)$$

$$\frac{[\text{Zn}^{2+}]}{[\text{Cu}^{2+}]} = \frac{0.02}{0.50} = 0.04 = 4.0 \times 10^{-2}$$

$$\log_{10}(0.04) = -1.3979$$

$$E_{\text{cell}} = 1.100 - \left(\frac{0.0591}{2}\right) \times (-1.3979) = 1.100 + 0.0413 = 1.1413\text{ V}$$

4. Maximum electrical work $W_{\max} = -\Delta G = nFE_{\text{cell}}$:

$$W_{\max} = 2 \times 96485 \times 1.1413 = 220237 \text{ J/mol} = 220.24 \text{ kJ/mol}$$